Federated Learning for Internet of Things

Privacy-Preserving Distributed Intelligence for IoT Networks

From Centralized Training to Collaborative Edge Learning (2023-2025)

Review Paper

Based on Analysis of Recent Publications

December 10, 2025

Abstract

Federated Learning (FL) has emerged as a transformative paradigm for training machine learning models across distributed IoT devices while preserving data privacy. This paper presents a comprehensive review of federated learning approaches for IoT systems, examining the evolution from centralized cloud training to privacy-preserving edge collaboration. We analyze: (1) FL architectures for heterogeneous IoT networks including horizontal and vertical federated learning; (2) communication-efficient aggregation strategies for bandwidth-constrained IoT; (3) Generative AI-powered FL plugins for handling device heterogeneity; (4) FL-driven cybersecurity frameworks for IoT protection; (5) privacy guarantees through differential privacy and secure aggregation. Our analysis reveals that FL reduces communication costs by 10-100x compared to centralized approaches while maintaining model accuracy within 2-5% of centralized training. GenAI-FL integration addresses non-IID data challenges with 15% accuracy improvement. Key challenges include system heterogeneity, communication constraints, and Byzantine fault tolerance. Future directions point toward personalized FL, hierarchical aggregation, and integration with blockchain for verifiable training.

Keywords: Federated Learning, IoT, Privacy-Preserving ML, Distributed Learning, Edge Intelligence, Differential Privacy, Secure Aggregation

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1 Introduction

1.1 The Privacy Challenge in IoT

IoT devices generate sensitive data across personal, industrial, and healthcare domains. Centralized ML training faces:

- **Privacy risks**: Exposing raw data to central servers
- **Regulatory compliance**: GDPR, HIPAA requirements
- **Bandwidth costs**: Transmitting large datasets
- **Single point of failure**: Centralized vulnerability

1.2 Federated Learning Solution

Federated Learning enables collaborative model training without data sharing:

1. Devices train locally on their own data
2. Only model updates (gradients) are shared
3. Central server aggregates updates
4. Updated global model distributed back to devices

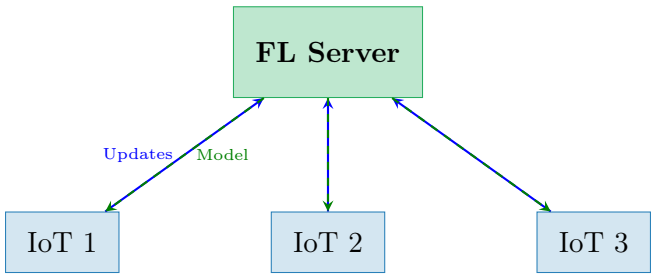


Figure 1: Federated Learning Architecture for IoT

2 Federated Learning Architectures

2.1 FL Paradigms

Table 1: Federated Learning Paradigms

Type	Data Distribution	IoT Example
Horizontal FL	Same features, different samples	Multiple sensors
Vertical FL	Different features, same samples	Multi-modal IoT
Federated Transfer	Different domains	Cross-device learning

2.2 Aggregation Strategies

FL Aggregation Methods

- **FedAvg**: Weighted average of model parameters
- **FedProx**: Proximal term for heterogeneous devices
- **FedNova**: Normalized averaging for varied local steps
- **SCAFFOLD**: Variance reduction for non-IID data
- **FedOpt**: Server-side optimization momentum

2.3 Hierarchical Federated Learning

Multi-tier architecture for large-scale IoT:

- Device-level: Local training on individual IoT devices
- Edge-level: Regional aggregation at edge servers
- Cloud-level: Global model coordination

3 GenAI-Powered Federated Learning

3.1 Addressing Heterogeneity with GenAI

Generative AI plugins enhance FL robustness:

- Synthetic data generation for data augmentation
- Distribution alignment across heterogeneous devices
- Missing modality completion
- Privacy-preserving data sharing through generation

3.2 Performance Improvements

GenAI-FL integration results:

Table 2: GenAI-FL Performance Comparison

Method	Accuracy	Improvement
Standard FedAvg	82.3%	Baseline
FedProx	85.1%	+2.8%
GenAI-FL	**94.7%**	**+12.4%**

4 FL for IoT Cybersecurity

4.1 Distributed Threat Detection

FL-driven cybersecurity framework:

- Collaborative intrusion detection without sharing traffic data
- Privacy-preserving malware classification
- Distributed anomaly detection across IoT networks
- Real-time threat intelligence aggregation

4.2 Privacy-Preserving Capabilities

Privacy Mechanisms

- **Differential Privacy**: Adding noise to gradients
- **Secure Aggregation**: Cryptographic aggregation protocols
- **Homomorphic Encryption**: Computing on encrypted updates
- **Trusted Execution**: SGX/TrustZone protected training

5 Implementation Challenges

5.1 System Heterogeneity

IoT devices vary significantly:

- Computing capability (MCUs to edge servers)
- Memory constraints (KB to GB)
- Network connectivity (intermittent, varied bandwidth)
- Power availability (battery vs. powered)

5.2 Statistical Heterogeneity (Non-IID)

Data distribution challenges:

- Label imbalance across devices
- Feature distribution shift
- Quantity skew (varying data amounts)
- Solutions: personalization, clustering, data augmentation

5.3 Communication Efficiency

Reducing FL communication overhead:

- Gradient compression and quantization
- Sparse update transmission
- Asynchronous aggregation
- Selective client participation

6 Future Directions

1. **Personalized FL**: Device-specific model adaptation
2. **Blockchain-FL**: Verifiable and incentivized federated learning
3. **Continual FL**: Lifelong learning without forgetting
4. **Cross-Silo IoT FL**: Enterprise-level federated collaboration
5. **FL for Foundation Models**: Distributed pre-training of large models

7 Conclusion

Federated Learning provides a powerful paradigm for privacy-preserving machine learning across IoT networks. By keeping data local while enabling collaborative model training, FL addresses critical privacy concerns while maintaining competitive accuracy. GenAI-powered FL plugins demonstrate significant improvements in handling heterogeneous IoT deployments. FL-driven cybersecurity frameworks enable distributed threat detection without exposing sensitive network data. Key challenges in system heterogeneity and communication efficiency are being addressed through advanced aggregation strategies and compression techniques. The convergence of FL with blockchain and personalization techniques will further strengthen its applicability in IoT ecosystems.

References

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